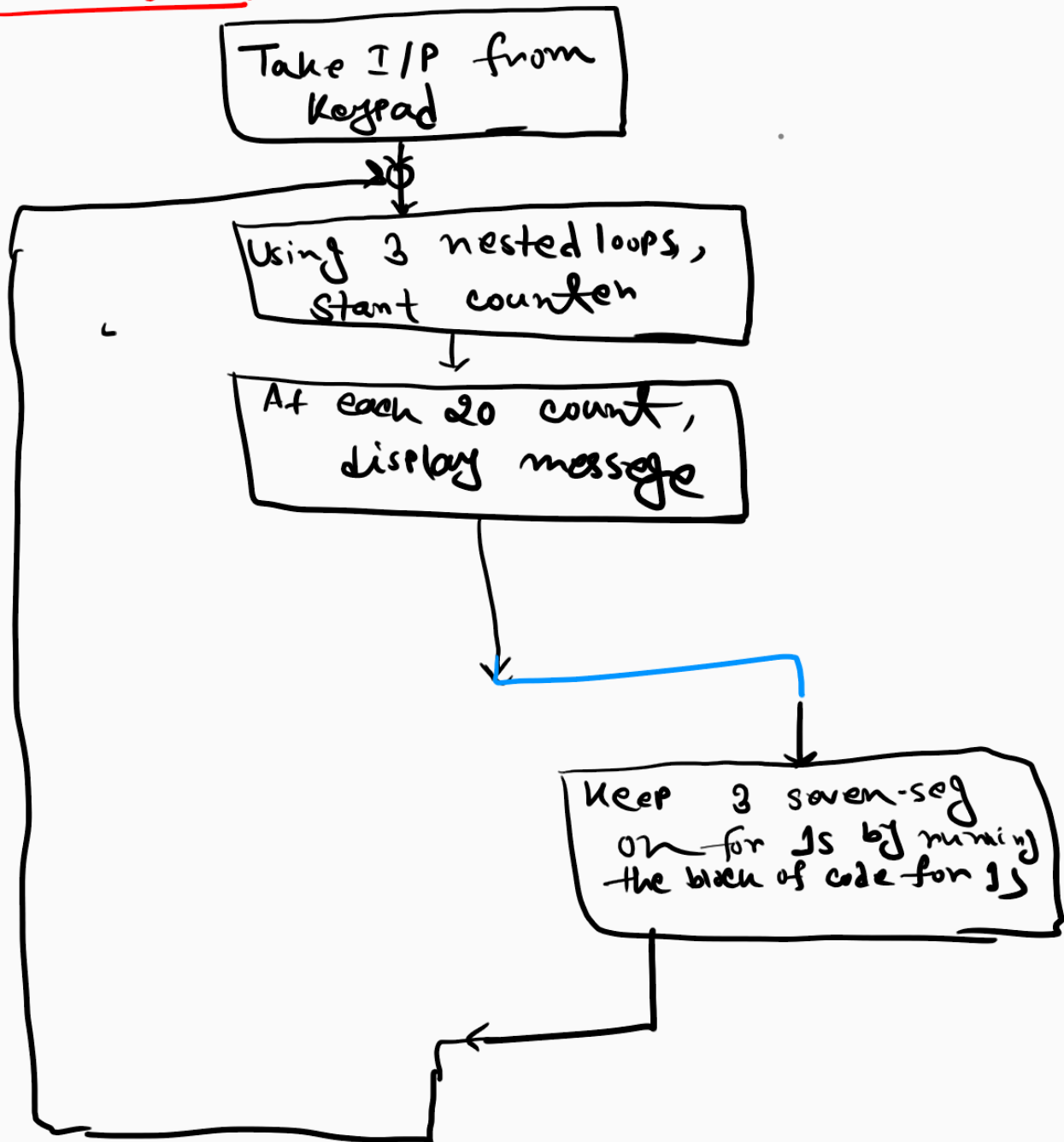


OVEN CEP

Workflow:



Stored number

Locations

70H, 71H, 72H

MSB → LSB

To check greater than 60:

R4	R3	R2
0	5	5
0	6	9
1	2	5
1	6	9

If R3 > 6:
True
Else if R4 > 0:
True
else: False

clock frequency calculation:

$$11 + \left\{ (22 - 12) \times \frac{200021125}{10^9} \right\}$$

$$= 13.00021125 \text{ MHz}$$

$$T = \frac{1}{\frac{1300021125}{12} \times 10^6} = 9.2306 \times 10^{-8} \text{ s}$$

$$= 9.231 \times 10^{-4} \text{ ms}$$

$$= 0.9231 \text{ ms}$$

Hardware configuration:

i) 3x4 keypad

↳ at port 3

↳ pin 4, 5, 6 are O/P rows

↳ pin 0, 1, 2, 3 are I/P columns

ii) LCD Display

↳ data pins at P1

↳ control pins are at pin 0, 1, 2 of port 0

iii) 7-segment display:

↳ connected at port 2

↳ multiplexed by pin 3, 4, 5 of port 0

iv) Status LED

↳ connected at pin 7 of port 3

↳ used a NOT gate

v) Buzzer:

↳ connected at pin 6, of port 0

↳ connected with a NOT gate as it operates on low signal

↳ initial low signal must be avoided

vii) Pull up resistor circuit

↳ Used a source, push button, 10k resistor, capacitor

↳ for reset purpose

viii) Additional pullup ckt for port 0:

↳ doesn't work like other ports in proteus

Software Part overview:

i) Initialization

↳ set up all names using EQU directive

↳ started MAIN away from Interrupt vector space

↳ Defined I/P, O/P Ports, initialized registers & pointer

↳ Enabled reset interrupt

ii) LCD initialization:

↳ Similar to lab

↳ printed a message to I/P time in 3 digit

iii) I/P through keypad:

↳ Similar to lab, just a 3x4 keypad

↳ I/P numbers are stored from memory 70H to 72H
↳ MSB ↳ LSB

↳ Function of C & S

iv) Counter loops :

- ↳ Made a 3 digit counter
- ↳ Used 3 nested loops
- ↳ When one of the digits at right becomes 0, it reduces the digit left to it and reset the digit at right to 9 and start again

↳ continues upto 000.

- ↳ also used a tracker (R5) to track how many numbers counted.

↳ When this is divisible by 20, that means it has gone 20s interval. So prints message

↳ otherwise just shows in 7SD.

- ↳ When ends, LED status switches
- ↳ activates buzzer.

v) Show_ZSG:

↳ activates one of the ZSG functions
then switches to other

↳ Entire block is run certain
times so that it takes
exactly 1 MC.

vi) Print_MESSAGE:

↳ first checks if time was
greater than 60 or not

↳ Then divides the tracken (RS)
and checks the dividend

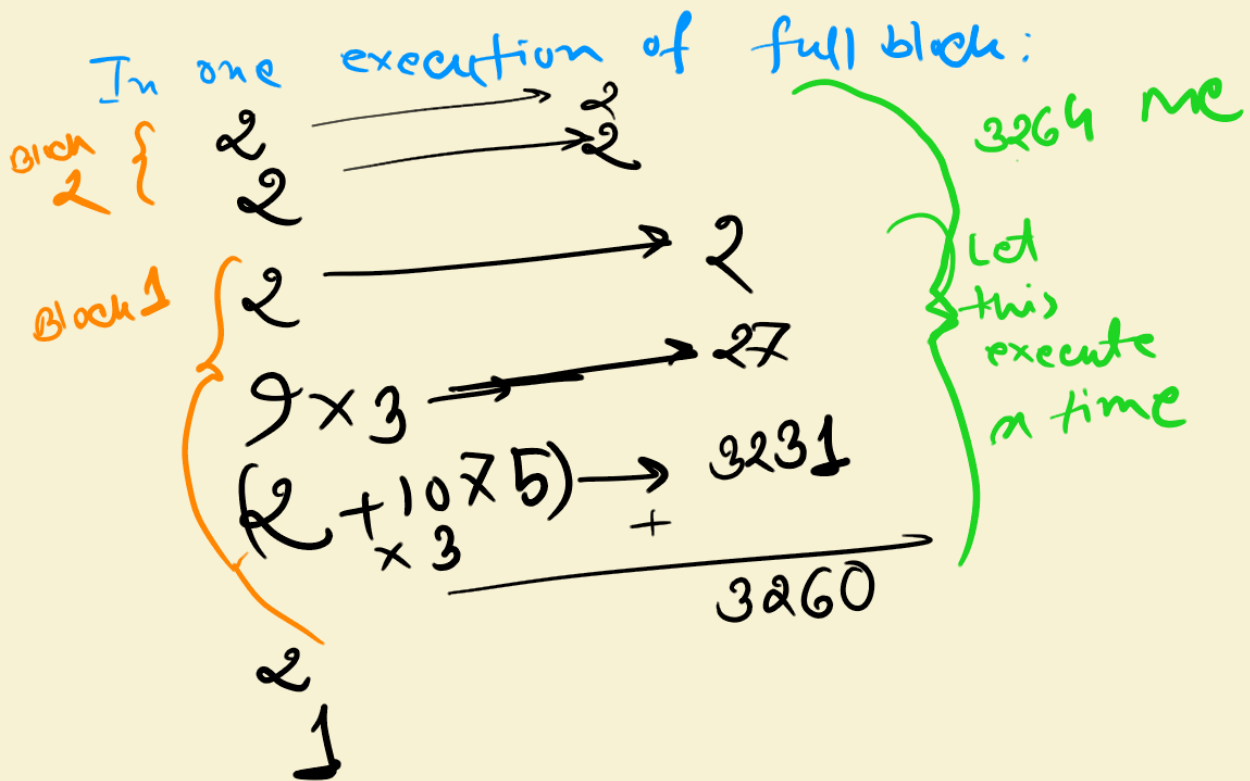
↳ If it is 0 → prints 1st msg

,, ,, , 1 → " 2nd "

!
so on

↳ Had to use multiple
intermediate labels to
avoid index out of range

Machine cycle & time calculation inside show-seg:



Now, $(2 + 3264 \times 2 + 1) \times 0.9231 \times 10^{-6} = 1$

✗ Can't be determined by this, as it is sometimes going to the print-message function also checking each time

Have to find by trial & error.